Ziad Ashraf

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AI Engineer with **3+** years of experience applying **Data Science**, **AI** and **Machine Learning** to solve real-world business problems. Hands-on experience in **predictive modeling**, **NLP, LLMs, Generative AI, RAG systems, vector databases**, **Speech AI**, **deep learning**, and end-to-end AI systems development. Skilled in taking AI problems from prototyping to production using **software engineering** principles, **MLOps**, and cloud-based deployment workflows, delivering independently in fully remote teams.

Experience

Elshayeb — Feb 2026 – Present
AI Engineer — *Alexandria | Remote*

Tools: Python, Gemini, Multimodal LLMs, Pydantic, Langfuse, Hybrid Search, BM25, Reranking, VAD, CTC Alignment

- Architected an **Egyptian Arabic speech recognition (ASR)** system used internally by the company's video editing team, combining **VAD**, **CTC forced alignment** and a **Multimodal LLM** to reach **8% macro WER** (3% CER) and cut manual transcription costs by **~66%**.
- Designed a two-stage multi-tenant SaaS lead retrieval system using **hybrid search (BM25 + dense embeddings)**, **reranking** and a custom relevance scoring algorithm, matching lead answers to tenant questionnaires and improving accuracy over baseline hard filters.
- Engineered an **AI automation** lead qualification service using **Gemini** to generate tenant-specific onboarding questionnaires, enforcing **Pydantic** structured outputs and context-constrained generation, with **Langfuse** for prompt versioning and cost optimization.

ISFP — Jul 2026 – Present
Data Engineer / BI Developer — *Alexandria | Remote*

Tools: Python, Oracle SQL, Alteryx, dbt, DuckDB, Parquet, Superset, Tableau Server, JWT, Row-Level Security

- Delivered operational data, BI and analytics solutions for **government port authorities** across the **Suez Canal Economic Zone**.
- Built **ETL pipelines** with **Alteryx**, **Oracle SQL**, **Python** and **dbt**, with incremental loading, data profiling checks and **data marts**.
- Deployed interactive **Tableau Server** KPI dashboards secured by **JWT**-based Connected Apps enforcing **row-level security (RLS)**.

HumaVolve — Jun 2026 – Sep 2026
AI Engineer Mentor — *Alexandria | Remote*

Tools: Python, FastAPI, Docker, RAG, FAISS, Prompt Engineering, Context Engineering, Chunking Strategies

- Led an **AI bootcamp** for fresh graduates, teaching them the full **production ML lifecycle** from prototype to cloud deployment.
- Delivered hands-on curriculum on **RAG systems** and **containerized FastAPI deployments** using Docker in real trainee projects.
- Mentored AI engineers via **architecture reviews**, **code quality audits** and **pipeline debugging** to deliver production-ready work.

Egyptian Armed Forces (R&D Lab) — Jun 2024 – Jan 2025
AI Engineer — *Cairo | Onsite*

Tools: Python, PyTorch, OpenCV, YOLO, UNet, BERT, Streamlit, ARIMA, ETS, Moving Averages

- Engineered healthcare deep learning pipelines with **YOLO** object detection and **UNet** segmentation to localize clinical anomalies.
- Developed **time-series anomaly detection** models (**ARIMA**, **ETS**, moving averages) on marine sensor data as AI PoC prototypes.
- Fine-tuned a **BERT** transformer for sentiment classification, winning **2nd Place** in an institutional AI Engineering Competition.

SimpliLearn Education Consulting Private Limited — Dec 2022 – Jun 2024
Data Scientist — *Alexandria | Remote*

Tools: Python, R, Power BI, Tableau, Feature Engineering, Hyperparameter Tuning

- Delivered data science solutions for academic and research clients, with **15+ client implementations** with a **95% acceptance rate**.
- Optimized ML models in **Python** and **R**, cutting baseline prediction error by **5–30%** via model tuning and feature engineering.
- Built interactive **Power BI** and **Tableau** dashboards with **100% on-time delivery**, helping drive a **25% rise in client retention**.

Freelancer.com — Jun 2022 – Apr 2024
Data Scientist — *Alexandria | Remote*

Tools: Python, AWS EC2, Linux, Cron Jobs, Spark, K-Means, PCA, UNet, Semantic Segmentation

- Deployed an ML system on **AWS EC2** with automated **cron-job** retraining in **Python** and **Linux**, achieving **100% error-free** deployment.
- Cut annotation time by **90%** using point-supervised semantic segmentation on aerial images with **UNet** and a partial focal loss.
- Segmented **100M+ customers** with **Spark K-Means**, reducing training time from 40 to 15 minutes by applying **PCA** reduction.

Projects

Real Estate Intelligence System

Tools: Python, FastAPI, Dagster, ClickHouse, Milvus, ONNX, Redis, MLflow, RAGAS, Docker, AWS

- Engineered a **Dagster** ETL and scraping pipeline on Docker, ingesting **10.9K+** Egyptian property listings into **ClickHouse** with automated **Pydantic** validation and data quality checks, achieving a **94.0% data quality score** for downstream semantic search and analytics.
- Architected a multi-stage **hybrid search** integrating LLM filter extraction, INT8-quantized multilingual-e5, **Milvus BM25** and an **ONNX bge-reranker**, raising **MRR from 0.059 to 0.412** (6.9x) and **NDCG@10 from 0.052 to 0.223** (4.3x) over a keyword-search baseline.
- Implemented a **FastAPI** serving layer with JWT/RBAC, sliding-window rate limiting and a two-tier cache (**Redis** exact + **Milvus** semantic), dropping P50 latency from **11.6s to <10ms**; benchmarked end to end with **RAGAS** and MLflow tracing at **82.1%** answer relevancy.

ContentLens AI: AI-Powered Marketing Document Platform

Tools: Python, LangGraph, LangChain, Langfuse, Ollama, Cohere, FastAPI, Tesseract, Pytest, AWS

- Architected a **LangGraph**-orchestrated multi-agent pipeline of **9 specialized agents** that analyze uploaded marketing documents in parallel (6 concurrent), reducing end-to-end **P90 latency by ~70%** to sub-3-minute response times under heavy concurrent load.
- Engineered **LangChain** agents with automatic **Ollama-to-Cohere failover** and retries, achieving **82.5% extraction quality** through rule-based validation combined with an **LLM-as-a-judge** evaluation step, keeping outputs reliable when the local Ollama model fails.
- Built a **FastAPI** backend ingesting PDF/DOCX/images/TXT with **Tesseract OCR**, multilingual translation, rate limiting and retries, traced with **Langfuse**, tested via PyTest and GitHub Actions CI, and deployed a cost-efficient **AWS EC2** PoC with no per-token API fees.

Black Friday: Retail Intelligence & ML-Powered Personalization Platform

Tools: Python, LangGraph, Gemini, Jev AI, LightGBM, ONNX Runtime, SHAP, PostgreSQL, Redis, FastAPI, Locust

- Architected a **LangGraph** agentic shopping bot with **Gemini** and **Jev AI** for personalized product recommendations, purchase guidance and **RAG**-powered catalog search, handling intent classification, off-topic routing, safety guardrails and adversarial blocking.
- Built a **LightGBM** personalized pricing engine combined with **Apriori** cross-sell rules, **Item2Vec** product similarity, 10-persona hierarchical clustering, MissForest imputation and **SHAP** explainability, with **Pandera** data validation and reproducible scikit-learn pipelines.
- Served models on **ONNX Runtime** (latency **14.2ms → 1.3ms**) with **Redis** multi-layer caching (semantic + exact match), batch inference and async **FastAPI** endpoints, keeping chatbot latency under **500ms** with **0.00% errors** at 50 concurrent users in **Locust** tests.

AI-Powered Mental Health Analysis System

Tools: Python, PyTorch, Hugging Face, LoRA, scikit-learn, TensorFlow, FastAPI, Airflow, MLflow, Docker, AWS

- Improved depression detection recall (TPR) by **31.9%** over **scikit-learn** sentiment baselines by fine-tuning **BERT** with **LoRA** via **Hugging Face** and **PyTorch**, enabling earlier detection of at-risk cases; also trained a TensorFlow CNN for emotion detection at 87% accuracy.
- Built a **FastAPI** microservice with **Docker** on **AWS** (ECR, EC2, RDS) and **GitHub Actions** CI/CD, cutting average API response time from **0.8s to 0.5s** by loading models once at startup and using async, non-blocking database operations and concurrent request handling.
- Orchestrated **3 ML training pipelines** (sentiment, depression detection and emotion CNN) with **Apache Airflow**, implementing data versioning and model registration with **S3**, **MLflow** and **WandB** for reproducible retraining and experiment tracking across models.

Skills

GenAI & Agents	LLMs, RAG, AI Agents, LangGraph, LangChain, CrewAI, Jev AI, OpenAI, LoRA/PEFT, Hugging Face
Search & Retrieval	Hybrid Search (BM25 + Dense), Reranking, Embeddings, Milvus, Qdrant, Semantic & Exact-Match Caching
Speech & Vision	ASR, VAD, CTC Forced Alignment, TTS, Multimodal LLMs, YOLO, UNet, OCR
ML Engineering	Python, SQL, PyTorch, TensorFlow, scikit-learn, LightGBM, XGBoost, ONNX Runtime, Time Series
MLOps & Cloud	FastAPI, Docker, Kubernetes, CI/CD, AWS, MLflow, Langfuse, RAGAS, Prometheus, Grafana, Loki
Data Engineering & BI	Airflow, Oracle, Spark, Kafka, dbt, ClickHouse, PostgreSQL, MongoDB, Alteryx, Tableau, Power BI

Education

Alexandria University — **Sep 2019 – Jun 2023**

Bachelor of Data Science and Computing — *Alexandria, Egypt*